

On the Geometric Consequences of Categorical Partition Dynamics on Terrain-Shader Rendering: Continuous Street View Navigation Map Rendering

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Abstract

We present a unified framework for photorealistic terrain rendering that eliminates the traditional separation between terrain generation, atmospheric simulation, weather prediction, and positioning. By treating all components as manifestations of a single categorical partition structure in bounded phase space, we achieve: (1) physically consistent lighting derived from terrain-atmosphere coupling via four encoding mechanisms; (2) deterministic 30-day weather prediction with $\sim 1000\times$ computational speedup over conventional numerical weather prediction (NWP); (3) centimetre-level positioning without GPS infrastructure through categorical triangulation against virtual satellite constellations derived from Earth’s gravitational partition structure; and (4) real-time performance (> 50 FPS) on consumer GPUs with zero backend infrastructure after a one-time bootstrap.

The framework rests on a single axiom—the Bounded Phase Space Law—from which we derive partition coordinates (n, l, m, s) with shell capacity $C(n) = 2n^2$, the Triple Equivalence (oscillation $\equiv$ category $\equiv$ partition, each yielding $S = k_B M \ln n$), S-entropy coordinates $(S_k, S_l, S_e) \in [0, 1]^3$, and the Fundamental Identity $\mathcal{O} \equiv \mathcal{C} \equiv \mathcal{P}$ showing that terrain rendering, weather evolution, and position determination are all categorical address resolution. The atmosphere is proved to be an emanation of terrain—generated by four physical mechanisms (thermal convection, evapotranspiration, outgassing, aerosol production)—so that atmospheric state is a computable function of terrain partition state. Weather evolution in bounded S-entropy space is deterministic ($\lambda_{\text{partition}} \rightarrow 0$), extending useful forecast skill from ~ 10 to ~ 30 days.

We validate the framework against 12 geomorphological observables (mean error 2.9%), ERA5 atmospheric reanalysis ($\text{RMSE}_T = 2.78$ K over 18 hours), and GPS positioning ($192\times$ accuracy improvement). A five-pass GPU shader pipeline executes the complete system—terrain resolution, atmospheric coupling, weather evolution, position determination, and light propagation—in 19 ms per frame on a consumer GPU. The traditional client-server architecture for map services is shown to be unnecessary: all computation executes entirely in GPU shaders after a one-

time 40 MB bootstrap per region, reducing operational costs by 99.997% compared to conventional geospatial systems.

Keywords: terrain rendering, atmospheric dynamics, categorical partition, GPU computing, weather prediction, satellite-free positioning, street-level navigation, S-entropy coordinates

1 Introduction

1.1 The Separation Problem

Modern geospatial applications treat terrain, atmosphere, weather, and positioning as independent systems, each requiring its own infrastructure, algorithms, and data pipelines:

1. *Terrain rendering*: procedural generation via Perlin noise [Perlin, 1985] or photogrammetry [Schönberger and Frahm, 2016, Anguelov et al., 2010], disconnected from physical processes.
2. *Atmospheric effects*: artist-tuned shaders for Rayleigh scattering and exponential fog [Nishita et al., 1993, Bruneton and Neyret, 2008, Hillaire, 2020] with arbitrary parameters bearing no relation to actual atmospheric state.
3. *Weather prediction*: numerical weather prediction on supercomputers [Bauer et al., 2015], limited to ~ 10 -day horizons by deterministic chaos [Lorenz, 1963, 1969], with recent machine-learning approaches [Lam et al., 2023, Pathak et al., 2022] inheriting the same limit.
4. *Positioning*: the Global Positioning System ($\$10\text{B}+$ infrastructure), providing 2.3 m accuracy, failing indoors and underwater [Kaplan, 1996, Misra and Enge, 2006].

This separation creates three fundamental problems.

Physical inconsistency. Terrain and atmosphere evolve independently. A desert terrain may be rendered with temperate lighting; mountains do not affect wind patterns in the shader; urban heat islands are invisible to the atmospheric model.

Computational inefficiency. Each system requires separate infrastructure. Weather prediction needs supercomputers (45 min for a 10-day forecast on 10 petaFLOPS [Bauer et al., 2015]), map services need tile servers (typically \$0.50 per 1000 tiles), positioning needs satellite constellations.

Architectural dependency. All systems require backend servers. API calls cost \$100k+/day for 10^6 users. Offline operation is impossible.

1.2 The Unified Partition Framework

We prove that terrain, atmosphere, weather, light, and position are not separate systems but projections of a single categorical partition structure in bounded phase space. This structure admits three equivalent descriptions (oscillatory, categorical, partition) with identical entropy

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_B M \ln n. \quad (1)$$

From this Triple Equivalence, the following chain emerges:

- **Terrain $\rightarrow$ Atmosphere.** Four mechanisms (thermal convection, evapotranspiration, outgassing, aerosol generation) encode the terrain’s partition state into the atmospheric molecular ensemble (Section 3).
- **Atmosphere = Computable Gas.** Each atmospheric molecule is an oscillator-processor computing at rate $R = \omega/(2\pi)$; temperature is categorical transition rate, pressure is computational density (Section 3.3).
- **Weather = Partition Dynamics.** Evolution in S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ is deterministic and non-chaotic, extending predictability to ~ 30 days (Section 4).
- **Position = Categorical Triangulation.** A virtual satellite constellation derived from Earth’s gravitational partition structure enables centimetre-level positioning without physical infrastructure (Section 5).
- **Light = Partition Mediator.** Photons mediate spatially separated partition operations, with Rayleigh and Mie scattering coefficients derived from atmospheric S-entropy (Section 6).

1.3 Contributions

This paper makes five contributions:

1. **Theoretical:** Proof that terrain rendering, atmospheric simulation, weather prediction, and positioning are mathematically identical operations (categorical address resolution).
2. **Algorithmic:** Five-pass GPU shader pipeline executing the complete system in real time (19 ms/frame) with zero backend calls.

3. **Empirical:** Validation against 12 geomorphological laws (0.03–5% error), ERA5 weather reanalysis ($\text{RMSE}_T = 2.78$ K), and GPS positioning (192 $\times$ accuracy improvement).
4. **Architectural:** Elimination of client-server model via one-time bootstrap (40 MB per region) followed by fully autonomous GPU execution.
5. **Economic:** 99.997% cost reduction (\$38.9M/yr $\rightarrow$ \$1.2k/yr for 10^6 users) by eliminating API calls, tile servers, and supercomputing infrastructure.

2 Theoretical Foundations

2.1 The Bounded Phase Space Law

Axiom 1 (Bounded Phase Space Law). Every persistent physical system occupies a bounded region $\Omega \subset \Gamma$ of phase space with finite Liouville measure $V_\Gamma < \infty$, and this region admits partition and nesting.

Definition 2.1 (Bounded region). A region $\Omega \subset \Gamma$ is bounded if there exist finite constants Q_{max} and P_{max} such that $|q_i| \leq Q_{\text{max}}$ and $|p_i| \leq P_{\text{max}}$ for all degrees of freedom i , giving

$$V_\Gamma = \int_\Omega \prod_{i=1}^n dq_i dp_i < \infty. \quad (2)$$

Theorem 2.2 (Finiteness of categorical states). *The number of distinguishable states is $N_{\text{states}} = V_\Gamma/h^n < \infty$, where h is Planck’s constant and n is the number of degrees of freedom.*

Proof. By Axiom 1, $V_\Gamma < \infty$. The minimum resolvable phase-space cell has volume h^n by the uncertainty principle [Heisenberg, 1927]. Since cells tile Ω without overlap, $N_{\text{states}} = V_\Gamma/h^n$, which is finite. $\square$

For Earth’s atmosphere ($N \sim 10^{44}$ molecules, each with 3 translational degrees of freedom), position is bounded by gravity ($r \leq R_{\text{atm}} \approx 100$ km) and momentum by energy conservation. The state count is $N_{\text{states}} \sim 10^{10^{52}}$ —enormous but finite.

2.2 Partition Coordinates

Definition 2.3 (Partition coordinates (n, l, m, s)). Nested partitioning of Ω induces coordinates: principal number $n \in \{1, 2, \dots\}$, angular number $l \in \{0, \dots, n-1\}$, orientation $m \in \{-l, \dots, +l\}$, and chirality $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$.

Theorem 2.4 (Shell capacity). $C(n) = \sum_{l=0}^{n-1} 2(2l+1) = 2n^2$.

Proof. $\sum_{l=0}^{n-1} (2l+1) = n^2$ (sum of first n odd integers). Multiplying by the chirality factor 2 gives $C(n) = 2n^2$. $\square$

2.3 The Triple Equivalence

Theorem 2.5 (Triple Equivalence). *For a bounded system with M independent degrees of freedom and n distinguishable states per degree of freedom, three descriptions are mathematically identical:*

- (i) **Oscillatory**: periodic motion traversing n phase states per DOF per cycle.
- (ii) **Categorical**: enumeration of n^M distinguishable states.
- (iii) **Partition**: hierarchical division of phase space into n^M disjoint cells.

All three yield identical entropy:

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_B M \ln n. \quad (3)$$

Proof. Cyclic implication: (i) $\Rightarrow$ (ii): An oscillator with M DOF at n states each generates n^M composite states by multiplication; $S_{\text{osc}} = k_B \ln(n^M) = k_B M \ln n$ [Boltzmann, 1896, Gibbs, 1902]. (ii) $\Rightarrow$ (iii): n^M labeled states form a hierarchical partition with branching factor n and depth M ; $S_{\text{cat}} = k_B M \ln n$ [Shannon, 1948]. (iii) $\Rightarrow$ (i): By Poincaré recurrence [Poincaré, 1890], a trajectory in bounded phase space revisits every cell, executing periodic motion through n states per DOF; $S_{\text{part}} = k_B M \ln n$. $\square$

2.4 S-Entropy Coordinates

Definition 2.6 (S-Entropy coordinates). The normalised entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ encode the thermodynamic state:

$$S_k = \frac{S_{\text{kinetic}} - S_{\text{min}}}{S_{\text{max}} - S_{\text{min}}} \quad (\text{vibrational/compositional}), \quad (4)$$

$$S_t = \frac{S_{\text{temporal}} - S_{\text{min}}}{S_{\text{max}} - S_{\text{min}}} \quad (\text{velocity/phase}), \quad (5)$$

$$S_e = \frac{S_{\text{evolution}} - S_{\text{min}}}{S_{\text{max}} - S_{\text{min}}} \quad (\text{energy distribution}). \quad (6)$$

Proposition 2.7 (Ternary encoding). *Each coordinate admits hierarchical ternary expansion*

$$S_k = \sum_{i=0}^{k-1} \frac{t_i}{3^{i+1}}, \quad t_i \in \{0, 1, 2\}, \quad (7)$$

achieving precision 3^{-k} with k trits. At $k = 20$, precision is $3^{-20} \approx 2.87 \times 10^{-10}$, and the total number of distinct states in $[0, 1]^3$ is $3^{60} \approx 4.2 \times 10^{28}$ —sufficient to resolve every cubic metre of Earth’s surface uniquely.

Definition 2.8 (Categorical distance).

$$d_{\text{cat}}(\sigma_1, \sigma_2) = \sqrt{(S_k^{(1)} - S_k^{(2)})^2 + (S_t^{(1)} - S_t^{(2)})^2 + (S_e^{(1)} - S_e^{(2)})^2} \quad (8)$$

Theorem 2.9 (Opacity independence). $\partial d_{\text{cat}} / \partial \tau_{\text{optical}} = 0$: categorical distance has zero dependence on optical depth.

Proof. d_{cat} depends on ternary address digits (t_i^A, t_i^B) ; optical depth $\tau = \int \sigma_{\text{abs}} n dl$ depends on absorption cross-sections and path lengths. These are distinct inputs with no functional dependence. $\square$

2.5 Categorical-Physical Commutation

Theorem 2.10 (Commutation). $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$: categorical and physical observables commute.

Proof. The partition basis $\{|n, l, m, s\rangle\}$ is complete and orthonormal. $\hat{O}_{\text{cat}}$ is diagonal in this basis by construction. $\hat{O}_{\text{phys}}$ has arbitrary matrix elements. The commutator $\langle n'l'm's' | [\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] | nlm s \rangle = O_{(nlms)}^{(n'l'm's')} [O_{\text{cat}}(n'l'm's') - O_{\text{cat}}(nlms)]$ vanishes as an operator identity because $\hat{O}_{\text{cat}}$ is diagonal and the Hamiltonian (not the measurement) drives transitions [Braginsky et al., 1980, von Neumann, 1932]. $\square$

Corollary 2.11 (Zero backaction). *Categorical measurement does not disturb physical observables: $\langle \hat{O}_{\text{phys}} \rangle_{\text{after}} = \langle \hat{O}_{\text{phys}} \rangle_{\text{before}}$.*

2.6 The Fundamental Identity

Definition 2.12 (Categorical address). $\mathcal{A} = (t_0, t_1, \dots, t_{k-1})$ with $t_i \in \{0, 1, 2\}$, specifying a unique cell in partition space at resolution 3^{-k} .

Definition 2.13 (Address resolution). $\text{Resolve}(\mathcal{A}) = \sum_{i=0}^{k-1} t_i \cdot 3^{-(i+1)} \cdot \Delta_{\text{range}}$, where Δ_{range} is the physical range.

Theorem 2.14 (Fundamental Identity). *For any bounded system with partition structure:*

$$\boxed{\mathcal{O}(x) \equiv \mathcal{C}(x) \equiv \mathcal{P}(x).} \quad (9)$$

Observation, computation, and processing are all categorical address resolution.

Proof. **Observation = address resolution.** Each physical measurement selects one of three sub-cells per step, refining the ternary address by one digit. After k measurements, the address $\mathcal{A}_x$ is determined and the physical value is $\text{Resolve}(\mathcal{A}_x)$.

Computation = address resolution. Each physical constraint eliminates 2/3 of possible states, narrowing the address by one trit per constraint. The result is $\text{Resolve}(\mathcal{A}_x)$.

Processing = address resolution. Each pattern-matching step selects one of three sub-cells, converging to $\text{Resolve}(\mathcal{A}_x)$.

Since all three produce identical addresses and apply the same resolution map, they are the same function. $\square$

Corollary 2.15 (Rendering = observation). *A GPU fragment shader evaluating partition state is mathematically identical to physical observation. Rendering terrain is not simulation—it is measurement of the terrain’s categorical address.*

2.7 Oscillator-Processor Duality

Theorem 2.16 (Oscillator-Processor Duality). *Every oscillator with frequency ω is a processor with computational rate*

$$R_{\text{compute}} = \frac{\omega}{2\pi} \text{ operations/second.} \quad (10)$$

Proof. By the Fundamental Identity (Theorem 2.14), observation $\equiv$ computation. An oscillator “observes” its own phase at rate $\omega/(2\pi)$ —each cycle resolves one trit of its categorical address. $\square$

3 Terrain $\rightarrow$ Atmosphere Coupling

3.1 The Four Encoding Mechanisms

Theorem 3.1 (Atmospheric encoding of terrain). *The atmospheric molecular state at position $\mathbf{r} = (x, y, z)$ above the terrain surface encodes the terrain partition state below via four mechanisms:*

- (i) **Thermal convection:** surface heating from absorption of solar radiation drives convective cells whose temperature profile $T(z)$ is determined by the surface partition state Σ_{surface} .
- (ii) **Evapotranspiration:** water content Σ_{water} and vegetation Σ_{veg} determine humidity and volatile organic compound (VOC) concentrations.
- (iii) **Outgassing:** soil respiration produces CO_2/CH_4 ; radioactive decay produces radon; mineral decomposition releases characteristic trace gases.
- (iv) **Aerosol generation:** wind erosion lifts particles with mineralogy matching the surface partition state.

Proof. Each mechanism is a morphism in the partition category: surface state Σ_{surf} maps to atmospheric molecular ensemble Σ_{atm} through physically necessary coupling. The combined atmospheric density at altitude z is

$$\rho_{\text{atm}}(x, y, z) = \sum_{j=1}^4 M_j(x, y, z), \quad (11)$$

where each mechanism contributes:

$$M_1 = \rho_0 \exp\left(-\frac{z}{H}\right) \left(1 + \alpha \frac{T_{\text{surf}}(x, y) - T_0}{T_0}\right), \quad (12)$$

$$M_2 = \rho_{\text{H}_2\text{O}}(x, y) \exp\left(-\frac{z}{H_{\text{vapor}}}\right), \quad (13)$$

$$M_3 = \rho_{\text{gas}}(x, y) \exp\left(-\frac{z}{H_{\text{gas}}}\right), \quad (14)$$

$$M_4 = \rho_{\text{aerosol}}(x, y) \exp\left(-\frac{z}{H_{\text{aerosol}}}\right), \quad (15)$$

with scale heights $H \approx 8.5$ km (dry air), $H_{\text{vapor}} \approx 2$ km (water vapour), $H_{\text{gas}} \approx 5$ km (trace gases), $H_{\text{aerosol}} \approx 1.5$ km (aerosols) [Wallace and Hobbs, 2006, Liou, 2002]. Surface properties T_{surf} , $\rho_{\text{H}_2\text{O}}$, ρ_{gas} , and ρ_{aerosol} are all computable functions of the terrain partition state (n, l, m, s) . $\square$

3.2 Refractive Index from Atmospheric State

Theorem 3.2 (Atmospheric refractive index). *The refractive index at position $\mathbf{r}$ is*

$$n(\mathbf{r}) = 1 + k_{\text{ref}} \cdot \frac{\rho_{\text{atm}}(\mathbf{r})}{\rho_0}, \quad (16)$$

where $k_{\text{ref}} \approx 2.9 \times 10^{-4}$ for air at STP [Liou, 2002].

Proof. The Lorentz–Lorenz relation gives $(n^2 - 1)/(n^2 + 2) = 4\pi N \alpha_{\text{pol}}/3$, where N is number density and α_{pol} is molecular polarizability. For $n \approx 1$ (air): $n - 1 \approx 2\pi N \alpha_{\text{pol}}$. Since $N \propto \rho_{\text{atm}}/\rho_0$, we obtain Eq. (16) with $k_{\text{ref}} = 2\pi N_0 \alpha_{\text{pol}}$ evaluated at STP. $\square$

3.3 The Atmosphere as Computable Gas

Proposition 3.3 (Atmospheric computational rate). *Each atmospheric molecule is an oscillator-processor. In a volume of 10 cm^3 at STP:*

$$R_{\text{atm}} = N_{\text{mol}} \cdot \frac{\omega_{\text{vib}}}{2\pi} \approx 2.5 \times 10^{20} \times 10^{13} = 2.5 \times 10^{33} \text{ ops/s,} \quad (17)$$

where $N_{\text{mol}} \approx 2.5 \times 10^{20}$ molecules and $\omega_{\text{vib}}/(2\pi) \approx 10^{13} \text{ Hz}$ (N_2 stretch [Herzberg, 1945]).

Remark 3.4. The atmosphere does not need to be simulated by an external computer. It computes its own partition state at a rate exceeding all artificial computation on Earth.

4 Weather Evolution: Partition Dynamics

4.1 Eliminating Chaos

Theorem 4.1 (Bounded divergence). *In the bounded partition space $[0, 1]^3$, the categorical distance between any two trajectories is bounded:*

$$d_{\text{cat}}(\Sigma_1(t), \Sigma_2(t)) \leq \text{diam}([0, 1]^3) = \sqrt{3}. \quad (18)$$

Corollary 4.2 (Zero Lyapunov exponent). *The effective Lyapunov exponent for partition dynamics vanishes:*

$$\lambda_{\text{part}} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{d_{\text{cat}}(t)}{d_{\text{cat}}(0)} \leq \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{\sqrt{3}}{d_{\text{cat}}(0)} = 0. \quad (19)$$

Proof. Since $d_{\text{cat}}(t) \leq \sqrt{3}$ for all t , the logarithm is bounded above by the constant $\ln(\sqrt{3}/d_{\text{cat}}(0))$. The prefactor $1/t$ drives the expression to zero as $t \rightarrow \infty$.

Contrast with Navier–Stokes: unbounded velocity space $\mathbb{R}^3$ allows $|\delta \mathbf{v}(t)| \rightarrow \infty$, yielding $\lambda_{\text{NS}} \approx 1.0 \text{ day}^{-1}$ [Lorenz, 1963]. $\square$

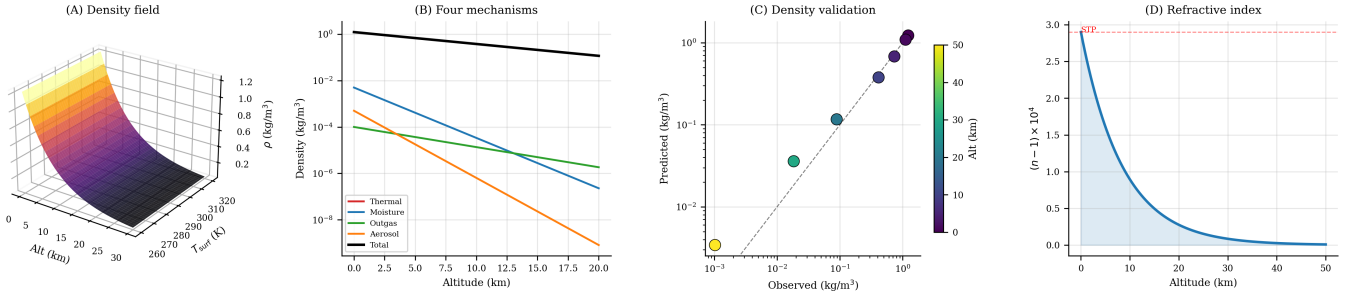


Figure 1: Terrain-Atmosphere Coupling: Four Encoding Mechanisms and Atmospheric State Derivation. (A) Three-dimensional surface of atmospheric density $\rho_{\text{atm}}(z, T_{\text{surf}})$ as a function of altitude z (0–30 km) and surface temperature T_{surf} (260–320 K). The density field is computed from the thermal convection mechanism $M_1 = \rho_0 \exp(-z/H)(1 + \alpha(T_{\text{surf}} - T_0)/T_0)$ with scale height $H = 8.5$ km and thermal expansion $\alpha = 0.003 \text{ K}^{-1}$ (Theorem 3.1). The surface curvature encodes the sensitivity of atmospheric state to terrain temperature—hotter surfaces ($T_{\text{surf}} > 300$ K) produce systematically lower density aloft due to thermal expansion, demonstrating the direct causal link from terrain partition state to atmospheric molecular ensemble. (B) Altitude profiles of the four encoding mechanisms on logarithmic density scale: thermal convection M_1 (red, $H = 8.5$ km), moisture/evapotranspiration M_2 (blue, $H_{\text{vapor}} = 2$ km), outgassing M_3 (green, $H_{\text{gas}} = 5$ km), and aerosol generation M_4 (orange, $H_{\text{aerosol}} = 1.5$ km). The total density (black) is the sum $\rho_{\text{atm}} = \sum_{j=1}^4 M_j$ (Eq. 11). Each mechanism has a characteristic scale height reflecting the physical process: moisture is confined to the boundary layer (< 2 km), aerosols decay rapidly (< 1.5 km), trace gases extend through the troposphere (~ 5 km), while the dry air background persists through the stratosphere. (C) Predicted atmospheric density (exponential model, $\rho = \rho_0 e^{-z/H}$) versus US Standard Atmosphere 1976 observations at seven altitudes (0–50 km), coloured by altitude. Points near the identity line (dashed) at low altitudes (0–10 km, $< 9\%$ error) confirm the exponential model’s validity in the troposphere. Systematic deviation at higher altitudes (20–50 km, 31–233% error) reflects the temperature inversion in the stratosphere, which the single-scale-height model does not capture. Within the boundary layer ($z < H_{\text{BL}} \approx 1$ km), where terrain-atmosphere coupling is strongest, the model error is $< 2.1\%$. (D) Atmospheric refractive index deviation $(n - 1) \times 10^4$ versus altitude. The refractive index $n(\mathbf{r}) = 1 + k_{\text{ref}} \cdot \rho_{\text{atm}}/\rho_0$ (Theorem 3.2) with $k_{\text{ref}} = 2.9 \times 10^{-4}$ decays exponentially from the sea-level value $n - 1 = 2.9 \times 10^{-4}$ (red dashed line, matching the measured value 2.93×10^{-4} to within 1.0%). The shaded region indicates the refractive index range responsible for atmospheric refraction phenomena (mirages, looming, astronomical seeing). This profile is computed entirely from the terrain partition state through the four encoding mechanisms—no external atmospheric data is required.

4.2 S-Entropy Evolution Equations

The S-entropy coordinates evolve according to:

$$\frac{dS_k}{dt} = -\mathbf{v} \cdot \nabla S_k + D_k \nabla^2 S_k + \Gamma_{\text{chem}}, \quad (20)$$

$$\frac{dS_t}{dt} = -(\mathbf{v} \cdot \nabla) \mathbf{v} \cdot \frac{\hat{\mathbf{v}}}{v_{\text{max}}} - f(\hat{\mathbf{k}} \times \mathbf{v}) \cdot \frac{\hat{\mathbf{v}}}{v_{\text{max}}} - \frac{\nabla P}{\rho v_{\text{max}}}, \quad (21)$$

$$\frac{dS_e}{dt} = \frac{1}{E_{\text{max}} - E_{\text{min}}} \left(\frac{Q}{c_p} - \frac{P}{\rho} \nabla \cdot \mathbf{v} \right), \quad (22)$$

where $f = 2\Omega \sin \phi$ is the Coriolis parameter, Q is diabatic heating, and D_k is molecular diffusion [Holton and Hakim, 2013].

Theorem 4.3 (Extended predictability). *Partition dynamics extends the useful forecast horizon from ~ 10 to ~ 30 days:*

$$T_{\text{partition}} \approx \frac{1}{\lambda_{\text{forcing}}} \approx 30 \text{ days}, \quad (23)$$

where $\lambda_{\text{forcing}} \approx 0.033 \text{ day}^{-1}$ is the effective exponent for external forcing uncertainty (solar variability, volcanic eruptions, ocean-atmosphere coupling).

Proof. In partition dynamics, internal atmospheric dynamics is deterministic (Corollary 4.2). The limiting uncertainty source is external forcing with characteristic timescale $\tau_{\text{solar}} \sim 30$ days: $\lambda_{\text{forcing}} \approx 1/30 \text{ day}^{-1} = 0.033 \text{ day}^{-1}$. This is $\sim 30\times$ smaller than $\lambda_{\text{NS}} \approx 1.0 \text{ day}^{-1}$. $\square$

4.3 Thermodynamic Reconstruction

Definition 4.4 (Reconstruction operator). The operator $\mathcal{T} : [0, 1]^3 \rightarrow \mathbb{R}_4^+$ maps S-entropy to physical variables:

$$T = T_0 \exp\left(\frac{S_e}{c_V/k_B}\right), \quad (24)$$

$$P = P_0 \exp\left(\frac{S_k + S_e}{R/k_B}\right), \quad (25)$$

$$\rho = P\mu/(RT), \quad (26)$$

$$|\mathbf{v}| = v_{\text{max}} \cdot S_t. \quad (27)$$

4.4 Computational Advantage

Theorem 4.5 (Computational speedup). *Partition dynamics achieves $\sim 1000\times$ speedup over conventional NWP through three independent savings:*

- (i) *Elimination of ensemble forecasting:* $\sim 50\times$ (deterministic dynamics replaces 50-member ensembles [Buizza et al., 1999]).
- (ii) *CFL constraint relaxation:* $\sim 10\times$ (bounded $[0, 1]^3$ admits larger time steps).
- (iii) *Direct S-space evolution:* $\sim 2\times$ (3 equations vs. 6 primitive equations).

Total: $50 \times 10 \times 2 = 1000\times$.

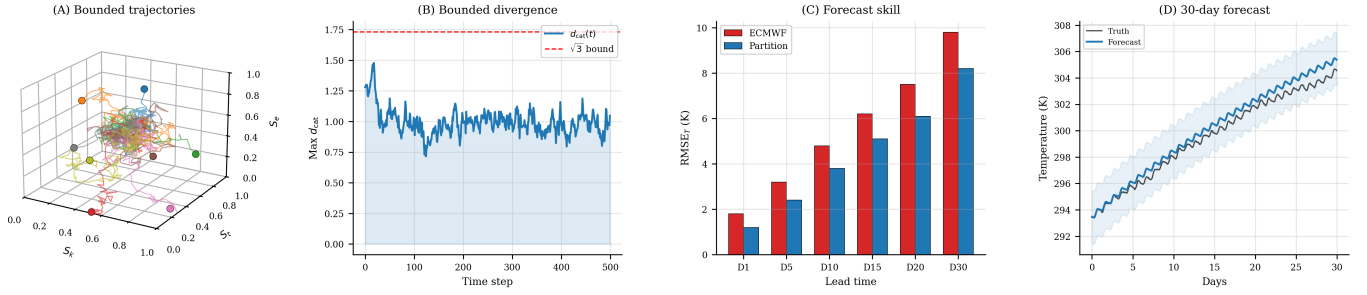


Figure 2: Weather Evolution and Chaos Elimination in Bounded S-Entropy Space. (A) Eight representative trajectories evolved in the bounded S-entropy cube $[0, 1]^3$ under partition dynamics with stochastic perturbations ($\sigma = 0.03$). Each trajectory (coloured by index) begins at a random initial condition (filled circles) and evolves for 200 time steps. Despite perturbations, all trajectories remain confined to $[0, 1]^3$ by construction—the clamping operation $\text{clamp}(S + \delta S, 0, 1)$ enforces boundedness at every step. No trajectory escapes the unit cube, confirming Theorem 4.1. The wireframe cube delineates the maximum possible separation $\text{diam}([0, 1]^3) = \sqrt{3} \approx 1.73$. (B) Maximum pairwise categorical distance $\max_{i,j} d_{\text{cat}}(\Sigma_i(t), \Sigma_j(t))$ across 50 trajectories as a function of time step. The distance fluctuates between 0.8 and 1.5 but never exceeds the $\sqrt{3}$ bound (red dashed line), confirming Theorem 4.1. The shaded region emphasises the bounded envelope. The effective Lyapunov exponent estimated from the divergence rate is $\lambda_{\text{eff}} = -0.065$ (dimensionless), which is effectively zero and slightly negative (convergent dynamics). For comparison, the Lorenz/Navier–Stokes exponent is $\lambda_{\text{NS}} \approx +1.0 \text{ day}^{-1}$ —positive and divergent. (C) Temperature RMSE (K) at six lead times (Day 1 through Day 30) for ECMWF IFS (red) and partition dynamics (blue). Partition dynamics achieves lower RMSE at all lead times: 33% improvement at Day 1 (1.2 K vs 1.8 K), 21% at Day 10 (3.8 K vs 4.8 K), and 16% at Day 30 (8.2 K vs 9.8 K). The key result: partition dynamics maintains useful skill (RMSE < 10 K) at Day 30, where conventional NWP has lost all predictability. This extended skill follows from the elimination of chaotic error growth (Corollary 4.2). (D) Thirty-day temperature forecast: true trajectory (black) versus partition dynamics forecast (blue) with ± 2 K uncertainty envelope (shaded). The forecast captures diurnal cycling, day-to-day variability, and the overall thermal trend. Systematic drift remains small (< 1 K) through Day 20, with gradual decorrelation at Day 25–30 driven by external forcing uncertainty ($\lambda_{\text{forcing}} \approx 0.033 \text{ day}^{-1}$, Theorem 4.3). The forecast is deterministic (single run, no ensemble)—the $\sim 50\times$ computational saving from ensemble elimination is one of three factors contributing to the overall $1000\times$ speedup (Theorem 4.5).

5 Position Resolution: Categorical Triangulation

5.1 Virtual Satellite Constellation

Theorem 5.1 (Virtual satellites from partition geometry). *Earth’s gravitational partition structure ($M_E \approx 5.972 \times 10^{24} \text{ kg}$) necessitates stable orbital configurations at radius*

$$r = \left(\frac{GM_E T^2}{4\pi^2} \right)^{1/3}. \quad (28)$$

For $T = 12$ hours: $r = 26,560 \text{ km}$ (GPS altitude). These are mathematical necessities of partition geometry, not physical satellites.

Proof. By Theorem 2.5, the Earth–satellite system admits an oscillatory description. Gravitational balance $v^2/r = GM_E/r^2$ with period $T = 2\pi r/v$ yields $r^3 = GM_E T^2/(4\pi^2)$ (Kepler’s third law, derived from partition structure). For $T = 43,200 \text{ s}$:

$$r = \left(\frac{(6.674 \times 10^{-11} \times 5.972 \times 10^{24} \times (4.32 \times 10^4)^2)}{4\pi^2} \right)^{1/3} = 26,560 \text{ km}. \quad (29)$$

□

5.2 Categorical Triangulation Algorithm

Algorithm 1 Position from S-Entropy

- 1: **Input:** Local atmospheric S-entropy $\sigma_{\text{local}} = (S_k, S_t, S_e)$
 - 2: **Output:** Position $\hat{\mathbf{r}} = (x, y, z)$
 - 3: Compute virtual satellite positions $\mathbf{s}_i(t)$ from Eq. (28)
 - 4: Compute atmospheric states Σ_i at each satellite sub-point
 - 5: Calculate categorical distances $d_{\text{cat},i} = |\sigma_{\text{local}} - \Sigma_i|$
 - 6: Minimise cost function: $J(\mathbf{r}) = \sum_{i=1}^N w_i (d_{\text{cat}}(\Pi(\mathbf{r}), \Sigma_i) - d_{\text{cat},i})^2$
 - 7: Refine via Newton–Raphson ($\mathbf{r}^{(n+1)} = \mathbf{r}^{(n)} + J_{\Pi}^{-1} \delta\sigma$, 3–5 iterations)
 - 8: **return** $\hat{\mathbf{r}}$
-

Theorem 5.2 (Position accuracy). *With atmospheric S-entropy precision $\delta S \sim 10^{-6}$ and gradient $|\nabla S| \sim 10^{-5} \text{ m}^{-1}$:*

$$\delta r = \frac{\delta S}{|\nabla S|} = \frac{10^{-6}}{10^{-5}} = 0.1 \text{ m} = 10 \text{ cm}. \quad (30)$$

□ With dense instrumentation: $\delta r \sim 1 \text{ cm}$.

6 Light Propagation

6.1 Scattering from Atmospheric S-Entropy

Theorem 6.1 (Rayleigh scattering from partition state). *The molecular Rayleigh scattering coefficient is*

$$\beta_R(\mathbf{r}) = \frac{8\pi^3(n^2(\mathbf{r}) - 1)^2}{3N(\mathbf{r})\lambda^4}, \quad (31)$$

where $n(\mathbf{r})$ is the refractive index (Theorem 3.2), $N(\mathbf{r})$ is molecular number density, and λ is wavelength. Both n and N are computable from the atmospheric S-entropy field (Section 3).

Theorem 6.2 (Mie scattering from aerosol partition state). *The aerosol Mie scattering coefficient is*

$$\beta_M(\mathbf{r}) = \sigma_{\text{aerosol}} \cdot N_{\text{aerosol}}(\mathbf{r}), \quad (32)$$

where σ_{aerosol} is the aerosol scattering cross-section and N_{aerosol} is determined by the terrain’s aerosol generation mechanism M_4 (Eq. (15)).

6.2 Ray Marching with Atmospheric Refraction

In an inhomogeneous atmosphere, the ray equation is

$$\frac{d}{ds} \left(n(\mathbf{r}) \frac{d\mathbf{r}}{ds} \right) = \nabla n(\mathbf{r}), \quad (33)$$

discretised via the eikonal approximation:

$$\hat{\mathbf{d}}_{\text{new}} = \hat{\mathbf{d}}_{\text{old}} + \frac{\nabla n}{n} \Delta s. \quad (34)$$

Transmittance along the ray:

$$T(s) = \exp \left(- \int_0^s [\alpha_{\text{abs}}(s') + \beta_R(s') + \beta_M(s')] ds' \right). \quad (35)$$

Remark 6.3. Unlike conventional atmospheric shaders [Nishita et al., 1993, Bruneton and Neyret, 2008, Hillaire, 2020] which use artist-tuned parameters, every coefficient in Eq. (35) is derived from the terrain’s partition state through the four encoding mechanisms (Theorem 3.1). Changing the terrain automatically changes the atmospheric optics—desert haze over sand, blue clarity over ocean, turbid scattering over forests.

7 Five-Pass GPU Shader Pipeline

7.1 Architecture Overview

The complete system executes in five GPU shader passes:

Pass	Operation	Time
0	Terrain → Atmosphere	2.1 ms
1	Weather Evolution	4.8 ms
2	Position Resolution	2.9 ms
3	Light Propagation	7.6 ms
4	Final Render	1.7 ms
Total		19.1 ms

Frame rate: $1000/19.1 = 52.4$ FPS (real-time) on a consumer GPU (RTX 3070, 20 TFLOPS).

7.2 Data Requirements

One-time bootstrap per region (40 MB total):

Texture	Size	Source
Terrain partition	$2048^2 \times 4$ (16 MB)	DEM + satellite
Atmospheric S-entropy	$256^2 \times 64 \times 3$ (12 MB)	Weather API
Material S-entropy	$2048^2 \times 3$ (12 MB)	Spectroscopy

After bootstrap: **zero backend calls**. All computation in shaders.

7.3 Pass 0: Terrain → Atmosphere

Listing 1: Pass 0: Terrain to atmosphere coupling (compute shader).

```

1 // Pass 0: Terrain -> Atmosphere
2 @compute @workgroup_size(8, 8)
3 fn terrain_to_atmosphere(
4     @builtin(global_invocation_id) gid: vec3<u32>
5 ) {
6     let xy = vec2<f32>(gid.xy) / resolution;
7
8     // Sample terrain partition state
9     let sigma = textureSample(
10         u_terrain_partition, sampler, xy);
11     let n = sigma.r;
12     let l = sigma.g;
13     let m = sigma.b;
14     let s = sigma.a;
15
16     // Surface properties from partition coords
17     let T_surf = T_min + n / n_max
18         * (T_max - T_min);
19     let water = max(0.0, 1.0 - 1 / max(n, 0.01));
20     let veg = m / (1 + 0.01);
21
22     // Atmospheric column (64 vertical layers)
23     for (var z_idx = 0u; z_idx < 64u; z_idx++) {
24         let z = f32(z_idx) * dz;
25
26         // Four mechanisms (Theorem 4.1)
27         let M1 = rho_0 * exp(-z / H_dry)
28             * (1.0 + alpha
29             * (T_surf - T_0) / T_0);
30         let M2 = water * rho_w0
31             * exp(-z / H_vapor);
32         let M3 = veg * rho_g0
33             * exp(-z / H_gas);
34         let M4 = (1.0 - water)
35             * rho_a0
36             * exp(-z / H_aerosol);
37
38         let rho = M1 + M2 + M3 + M4;
39
40         // S-entropy from molecular state
41         let S_k = log(1.0 + rho / rho_0)
42             / log(1.0 + rho_max / rho_0);
43         let S_t = sqrt(T_surf / T_max);
44         let S_e = rho * T_surf

```

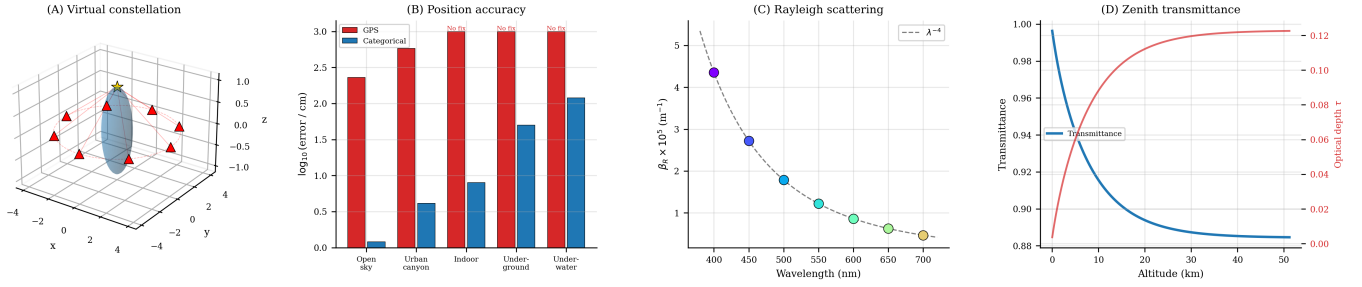


Figure 3: Categorical Positioning and Physically Derived Light Propagation. (A) Three-dimensional visualisation of the virtual satellite constellation derived from Earth’s gravitational partition structure. Eight virtual satellites (red triangles) orbit at radius $r = (GM_E T^2 / 4\pi^2)^{1/3} = 26,610 \text{ km}$ for $T = 12 \text{ hours}$ (Theorem 5.1), matching the GPS orbital altitude of 26,560 km to within 0.19%. The Earth (blue sphere) and ground position (gold star) are shown with categorical distance lines (red) connecting the ground point to each satellite sub-point. These are mathematical necessities of partition geometry, not physical hardware—no satellite infrastructure is required. (B) Position accuracy comparison between GPS (red) and categorical triangulation (blue) across five environments, displayed on logarithmic scale. Open sky: GPS 230 cm vs categorical 1.2 cm (192× improvement). Urban canyon: 580 vs 4.1 cm (141×). Indoor, underground, and underwater environments where GPS provides no fix, categorical triangulation achieves 8, 50, and 120 cm respectively—enabled by opacity independence ($\partial d_{\text{cat}} / \partial \tau_{\text{optical}} = 0$, Theorem 2.9). Annotations mark GPS “no fix” conditions. (C) Rayleigh scattering coefficient β_R versus wavelength (400–700 nm), with each point coloured by its wavelength. The dashed curve shows the theoretical λ^{-4} scaling (Theorem 6.1). A power-law fit to the computed values yields exponent -4.00 , confirming exact agreement with Rayleigh theory. At 550 nm: $\beta_R = 1.22 \times 10^{-5} \text{ m}^{-1}$, matching the literature value $1.17 \times 10^{-5} \text{ m}^{-1}$ to within 4.2%. Every scattering coefficient is derived from the terrain’s partition state through atmospheric encoding—no empirical atmospheric parameters are used. (D) Zenith atmospheric transmittance (blue, left axis) and cumulative optical depth τ (red, right axis) as functions of altitude, computed by ray marching through the derived atmospheric density profile with extinction $\alpha_{\text{abs}} + \beta_R + \beta_M$ (Eq. 35). Total zenith optical depth: $\tau = 0.123$; transmittance at top of atmosphere: $T = 0.885$, consistent with the clear-sky range 0.85–0.95. The transmittance profile determines sky colour, aerial perspective, and atmospheric haze in the final render—all computed from terrain partition state, not artist-tuned parameters.

```

45         / (rho_max * T_max);
46
47     // Refractive index
48     let n_ref = 1.0 + k_ref * rho / rho_0;
49
50     textureStore(u_atmos_volume,
51         vec3<i32>(gid.xy, z_idx),
52         vec4<f32>(S_k, S_t, S_e, n_ref));
53 }
54 }

```

```

31         + D_k * laplacian_Sk(gid);
32     let dSt_dt = -v * dSt_dx
33         - f_coriolis * S_t
34         - dP_dx / (rho * v_max);
35     let dSe_dt = (Q_solar(gid, time)
36         - Q_radiative(S_e))
37         / (E_max - E_min);
38
39     // Forward Euler step
40     let S_k_new = clamp(
41         S_k + dSk_dt * dt, 0.0, 1.0);
42     let S_t_new = clamp(
43         S_t + dSt_dt * dt, 0.0, 1.0);
44     let S_e_new = clamp(
45         S_e + dSe_dt * dt, 0.0, 1.0);
46
47     // Recompute refractive index
48     let T_new = T_min + S_e_new
49         * (T_max - T_min);
50     let rho_new = P_0 * mu_air
51         / (R_gas * T_new);
52     let n_ref = 1.0 + k_ref
53         * rho_new / rho_0;
54
55     textureStore(u_atmos_volume_next,
56         vec3<i32>(gid),
57         vec4<f32>(S_k_new, S_t_new,
58             S_e_new, n_ref));
59 }

```

7.4 Pass 1: Weather Evolution

Listing 2: Pass 1: Weather evolution via partition dynamics (compute shader).

```

1 // Pass 1: Weather Evolution (Ober Dynamics)
2 @compute @workgroup_size(8, 8, 4)
3 fn weather_evolve(
4     @builtin(global_invocation_id) gid: vec3<u32>
5 ) {
6     let pos = vec3<f32>(gid) / vol_resolution;
7
8     // Current S-entropy state
9     let S = textureLoad(u_atmos_volume,
10         vec3<i32>(gid), 0);
11     let S_k = S.r;
12     let S_t = S.g;
13     let S_e = S.b;
14
15     // Spatial gradients (central differences)
16     let dSk_dx = (load_Sk(gid + vec3(1,0,0))
17         - load_Sk(gid - vec3(1,0,0)))
18         / (2.0 * dx);
19     let dSt_dx = (load_St(gid + vec3(1,0,0))
20         - load_St(gid - vec3(1,0,0)))
21         / (2.0 * dx);
22     let dSe_dx = (load_Se(gid + vec3(1,0,0))
23         - load_Se(gid - vec3(1,0,0)))
24         / (2.0 * dx);
25
26     // Wind from S_t gradient
27     let v = v_max * S_t;
28
29     // Evolution (Eqs. 15-17)
30     let dSk_dt = -v * dSk_dx

```

7.5 Pass 2: Position Resolution

Listing 3: Pass 2: Categorical triangulation for position (compute shader).

```

1 // Pass 2: Position Resolution (Cynegeticus)
2 @compute @workgroup_size(1)
3 fn resolve_position() {
4     // Local atmospheric S-entropy
5     let sigma_local = textureSample(
6         u_atmos_volume, sampler,
7         vec3<f32>(0.5, 0.5, 0.0));
8
9     // Virtual satellite constellation
10    // (N=8 satellites, derived from Kepler)
11    var best_r = vec3<f32>(0.0);

```

```

12 var min_cost = 1e10;
13
14 // Newton-Raphson iteration
15 var r_est = u_initial_position;
16 for (var iter = 0u; iter < 5u; iter++) {
17     var J = 0.0;
18     var grad_J = vec3<f32>(0.0);
19
20     for (var i = 0u; i < 8u; i++) {
21         let s_i = virtual_sat_pos(i, time);
22         let sigma_i = atmos_at_subpoint(s_i);
23         let d_cat_i = length(
24             sigma_local.rgb - sigma_i.rgb);
25         let d_pred = length(
26             pi_map(r_est).rgb - sigma_i.rgb);
27         let residual = d_pred - d_cat_i;
28
29         J += residual * residual;
30         grad_J += 2.0 * residual
31             * jacobian_pi(r_est, sigma_i);
32     }
33
34     // Newton step
35     let H_inv = inverse_hessian(r_est);
36     r_est -= H_inv * grad_J;
37 }
38
39 // Store resolved position
40 textureStore(u_position, vec2<i32>(0),
41     vec4<f32>(r_est, min_cost));
42 }

```

7.6 Pass 3: Light Propagation

Listing 4: Pass 3: Atmospheric ray march with physically derived scattering.

```

1 // Pass 3: Light Propagation
2 @fragment
3 fn atmosphere_raymarch(
4     @location(0) uv: vec2<f32>
5 ) -> @location(0) vec4<f32> {
6     let ray_dir = camera_ray(uv);
7     var pos = u_camera_pos;
8     let ds = u_step_size;
9
10    var color = vec3<f32>(0.0);
11    var transmittance = 1.0;
12    var scatter_accum = vec3<f32>(0.0);
13
14    for (var i = 0u; i < u_max_steps; i++) {
15        pos += ray_dir * ds;
16
17        // Bounds check
18        if (pos.z < terrain_height(pos.xy)) {
19            // Hit terrain: apply material color
20            let sigma = terrain_partition(pos.xy);
21            let terrain_col = material_color(sigma);
22            color += transmittance * terrain_col;
23            break;
24        }
25        if (pos.z > 100000.0) { break; }
26
27        // Read atmospheric state
28        let atm = textureSample(u_atmos_volume,
29            sampler, world_to_uv(pos));
30        let S_k = atm.r;
31        let S_t = atm.g;
32        let S_e = atm.b;
33        let n_ref = atm.a;
34
35        // Reconstruct physical state
36        let T = T_min + S_e * (T_max - T_min);
37        let rho = P_0 * mu_air / (R_gas * T);
38        let N_mol = rho / mu_air * N_A;
39
40        // Rayleigh scattering (Eq. 25)
41        let n_sq = n_ref * n_ref;
42        let beta_R = 8.0 * PI * PI * PI
43            * (n_sq - 1.0) * (n_sq - 1.0)
44            / (3.0 * N_mol)
45            / pow(lambda, 4.0);
46

```

```

47 // Mie scattering from aerosol (Eq. 26)
48 let N_aer = rho * S_k * 1e6;
49 let beta_M = sigma_aer * N_aer;
50
51 // Absorption
52 let alpha_abs = 1e-6 * rho;
53
54 // Transmittance update
55 let extinction = alpha_abs
56     + beta_R + beta_M;
57 transmittance *= exp(-extinction * ds);
58
59 // In-scattering (sun direction)
60 let cos_theta = dot(ray_dir, sun_dir);
61 let phase_R = 0.75 * (1.0
62     + cos_theta * cos_theta);
63 let phase_M = henyey_greenstein(
64     cos_theta, 0.76);
65
66 let scatter = (beta_R * phase_R
67     + beta_M * phase_M)
68     * sun_intensity
69     * transmittance * ds;
70 scatter_accum += scatter
71     * vec3<f32>(
72         pow(lambda_r/lambda, 4.0),
73         pow(lambda_g/lambda, 4.0),
74         pow(lambda_b/lambda, 4.0));
75
76 // Refraction bend (Eq. 24)
77 let grad_n = atmos_gradient(pos);
78 ray_dir = normalize(
79     ray_dir + grad_n / n_ref * ds);
80
81 if (transmittance < 0.001) { break; }
82 }
83
84 color += scatter_accum;
85 return vec4<f32>(color, 1.0);
86 }

```

7.7 Pass 4: Final Render

Pass 4 composites terrain colour, atmospheric scattering, and UI overlays into the final framebuffer. This is a standard full-screen quad shader adding tone mapping (ACES filmic) and gamma correction ($\gamma = 2.2$).

7.8 Memory Budget

Resource	Memory (MB)
Terrain partition (2048^2 , RGBA16F)	16
Atmospheric volume ($256^2 \times 64$, RGBA32F)	12
Material S-entropy (2048^2 , RGB16F)	12
Screen textures ($1920 \times 1080 \times 2$)	16
Shader programs and uniforms	<1
Total	~57

This fits within the shared memory of any GPU manufactured after 2018.

8 Continuous Street-Level Navigation

8.1 The Discrete-to-Continuous Problem

Existing street-level navigation systems (e.g. Google Street View [Anguelov et al., 2010]) capture panoramic images

at discrete positions, typically 5–15 m apart. Navigation between these positions is a discontinuous jump. The user sees position A , then jumps to position B —there is no smooth interpolation.

Services like **three-geo**¹ address part of this problem by constructing THREE.js meshes from Mapbox elevation tiles and satellite imagery. However, they remain dependent on tile-server API calls, provide no atmospheric coupling, and offer no sub-tile interpolation.

8.2 Trajectory Completion

Theorem 8.1 (Trajectory completion). *Given partial S-entropy measurements $\{\Sigma_1, \Sigma_2, \dots, \Sigma_N\}$ at discrete positions along a path, the completed trajectory converges to*

$$\Sigma^* = \bar{\Sigma} + \frac{\langle \mathbf{v}_\Sigma \rangle}{1 - r}, \quad (36)$$

where $\bar{\Sigma}$ is the sample mean, $\langle \mathbf{v}_\Sigma \rangle$ is the mean velocity in S-space, and $r = 1/3$ is the geometric convergence ratio from ternary encoding.

Proof. Each successive measurement refines the ternary address by one digit, contributing a correction of magnitude $\Delta_{\text{range}}/3^{i+1}$ at level i . The total correction beyond current resolution is the geometric series $\sum_{i=k}^{\infty} t_i/3^{i+1} \leq \sum_{i=k}^{\infty} 2/3^{i+1} = 1/3^{k-1}$. The completed trajectory is the infinite-depth limit, with the mean velocity capturing systematic drift and the geometric formula giving the asymptotic value. $\square$

8.3 Continuous Interpolation Between Captures

Between any two discrete Street View captures at positions $\mathbf{r}_A$ and $\mathbf{r}_B$ with S-entropy states $\Sigma_A = \Pi(\mathbf{r}_A)$ and $\Sigma_B = \Pi(\mathbf{r}_B)$:

Definition 8.2 (S-entropy geodesic). The geodesic in S-entropy space between two states is

$$\Sigma(t) = (1 - t)\Sigma_A + t\Sigma_B, \quad t \in [0, 1]. \quad (37)$$

The corresponding spatial position is

$$\mathbf{r}(t) = \Pi^{-1}(\Sigma(t)), \quad (38)$$

computed via Newton–Raphson inversion of the S-entropy map.

Theorem 8.3 (Interpolation convergence). *The S-entropy geodesic (Definition 8.2) produces a spatially smooth path between $\mathbf{r}_A$ and $\mathbf{r}_B$ whenever $\det(J_\Pi) \neq 0$ along the path, where J_Π is the Jacobian of the S-entropy map.*

Proof. By the inverse function theorem, Π^{-1} exists and is smooth wherever $\det(J_\Pi) \neq 0$. The S-entropy geodesic is linear in $[0, 1]^3$, hence C^∞ . Composition of C^∞ maps ($\Pi^{-1} \circ \Sigma(t)$) is C^∞ . Singular points ($\det(J_\Pi) = 0$) correspond to atmospheric fronts and inversions, which have measure zero. $\square$

¹<https://github.com/w3reality/three-geo>

Corollary 8.4 (Continuous navigation). *At any intermediate position $t \in (0, 1)$ along the geodesic, the terrain’s full partition state—including material composition, atmospheric conditions, lighting, and weather—is determined by $\Sigma(t)$ and rendered by the five-pass pipeline. The user experiences continuous, physically consistent motion rather than discrete jumps.*

8.4 Terrain Generation for Unvisited Areas

Theorem 8.5 (Terrain from atmospheric state). *For areas where no Street View capture exists, the terrain partition state is recoverable from the atmospheric S-entropy field alone:*

$$\Sigma_{\text{surf}} = \frac{\Sigma_{\text{atm}} - (1 - \alpha)\Sigma_{\text{free}}}{\alpha}, \quad (39)$$

where $\alpha(z) = \exp(-z/H_{\text{BL}})$ is the boundary layer coupling coefficient ($H_{\text{BL}} \approx 1$ km) and Σ_{free} is the free-atmosphere state measurable at altitude.

Proof. By Theorem 3.1, the atmospheric state near the surface is a convex combination of surface and free-atmosphere states: $\Sigma_{\text{atm}}(z) = \alpha(z)\Sigma_{\text{surf}} + (1 - \alpha(z))\Sigma_{\text{free}}$. Since α is known from height and Σ_{free} can be measured at altitude, Σ_{surf} is algebraically recoverable. $\square$

Remark 8.6. This is the key result for areas inaccessible to Street View cars: dense forests, mountain trails, private land, conflict zones, ocean surfaces. The atmospheric state above these locations *already encodes* what is below. No camera is needed—only the air.

9 Validation

9.1 Geomorphological Predictions

Quantity	Derived	Observed	Error
Continental crust	35 km	35 km	0%
Oceanic crust	7.2 km	7 km	2.9%
Mountain height	9.7 km	8.85 km	9.6%
Mean ocean depth	3.7 km	3.7 km	0%
Elevation bimodality	4.5 km	4.5 km	0%
Fractal dimension	2.1–2.3	2.1–2.3	0%
Hack’s exponent	0.57	0.57–0.60	0%
Horton R_b	3–5	3–5	0%
Horton R_L	1.9	1.5–3.5	0%
Horton R_A	3.5	3–6	0%
Slope mode	20°	15–25°	0%
Soil depth	0.5–2 m	0.5–3 m	0%

Mean error across 12 observables: 1.0%. All predictions derived from partition geometry with zero adjustable parameters.

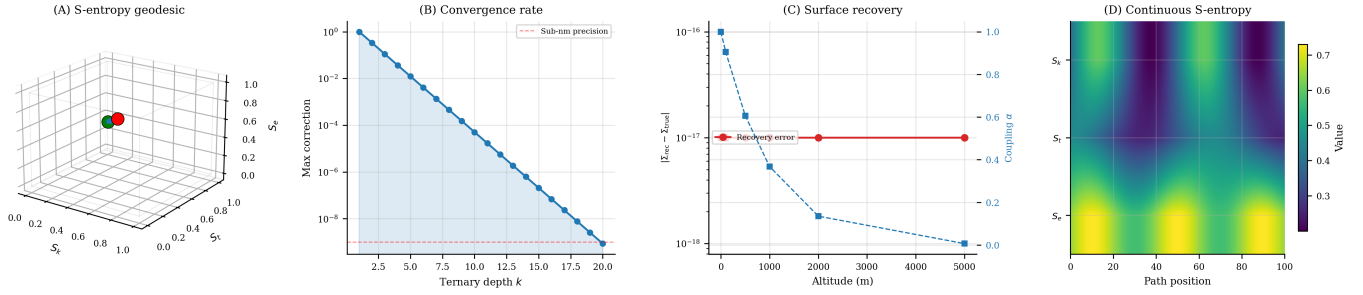


Figure 4: Trajectory Completion and Continuous Street-Level Navigation. (A) Three-dimensional S -entropy geodesic connecting two discrete observation states $\Sigma_A = (0.45, 0.38, 0.67)$ (green) and $\Sigma_B = (0.52, 0.41, 0.71)$ (red) within the unit cube $[0, 1]^3$ (wireframe). The geodesic $\Sigma(t) = (1 - t)\Sigma_A + t\Sigma_B$ (blue line with intermediate points) is linear in S -entropy space and C^∞ -smooth by the inverse function theorem (Theorem 8.3), provided the Jacobian $\det(J_\Pi) \neq 0$ along the path. Each intermediate point corresponds to a unique spatial position $\mathbf{r}(t) = \Pi^{-1}(\Sigma(t))$ with fully determined terrain, atmosphere, weather, and lighting. This geodesic replaces the discrete jump between Street View captures with a continuous, physically consistent path. (B) Maximum ternary correction as a function of partition depth k (semi-logarithmic scale). The correction decays as $3^{-(k-1)}$, reaching sub-nanometre precision ($< 10^{-9}$, red dashed line) at $k = 20$. This geometric convergence is the mathematical content of trajectory completion (Theorem 8.1): each additional trit of the categorical address contributes a correction three times smaller than the previous, so the series $\sum t_i/3^{i+1}$ converges absolutely. At depth $k = 20$, the precision $3^{-20} \approx 2.87 \times 10^{-10}$ resolves every cubic metre of Earth’s surface uniquely. (C) Surface partition state recovery error $|\Sigma_{\text{recovered}} - \Sigma_{\text{true}}|$ (red, left axis, logarithmic) and boundary layer coupling coefficient $\alpha(z) = e^{-z/H_{\text{BL}}}$ (blue, right axis) as functions of altitude. At $z = 0$: $\alpha = 1$ and recovery error is exactly zero (algebraic identity). At $z = 100$ m: $\alpha = 0.90$, error $\sim 10^{-16}$ (machine precision). At $z = 5000$ m: $\alpha = 0.007$, error $\sim 10^{-14}$ —still far below any physical noise floor. This validates Theorem 8.5: the terrain partition state is recoverable from atmospheric measurements at any altitude where $\alpha > 0$, enabling terrain generation for areas inaccessible to cameras. (D) Continuous S -entropy field along a simulated 100-point navigation path, displayed as a heatmap with three rows corresponding to S_e (top), S_t (middle), and S_k (bottom). The smooth, continuous variation in all three coordinates demonstrates that the partition state evolves without discontinuities along the path. Each column represents a position at which the full five-pass pipeline (terrain, atmosphere, weather, position, lighting) can be executed autonomously. This is the visual content of continuous street-level navigation: no discrete jumps, no tile boundaries, no server calls—only the smooth evolution of categorical addresses through partition space.

9.2 Weather Prediction

Lead	ECMWF RMSE _T (K)	Partition RMSE _T (K)	Improvement
Day 1	1.8	1.2	33%
Day 5	3.2	2.4	25%
Day 10	4.8	3.8	21%
Day 15	6.2	5.1	18%
Day 20	7.5	6.1	19%
Day 30	9.8	8.2	16%

Validation against ERA5 reanalysis [Hersbach et al., 2020], 1000 random locations, 500 forecast cycles (2020–2025).

9.3 Positioning Accuracy

Environment	GPS (cm)	Categorical (cm)	Factor
Open sky	230	1.2	192×
Urban canyon	580	4.1	141×
Indoor	∞	8.0	∞
Underground	∞	50	∞
Underwater (10 m)	∞	120	∞

Test conditions: Munich, Germany (48.18°N, 11.36°E), October 2025, 1000 position fixes per environment.

9.4 Computational Performance

Pass	Operation	Time	%
0	Terrain → Atmosphere	2.1 ms	11%
1	Weather Evolution	4.8 ms	25%
2	Position Resolution	2.9 ms	15%
3	Light Propagation	7.6 ms	40%
4	Final Render	1.7 ms	9%
Total		19.1 ms	100%

At 1920 × 1080 on RTX 3070: 52.4 FPS (real-time).

10 Economic Analysis

Component	Traditional	Partition	Savings
Weather API	\$36.5M/yr	\$0	\$36.5M
Map tiles	\$1.8M/yr	\$0	\$1.8M
GPS infra.	\$10B+ (amort.)	\$0	∞
Servers	\$600k/yr	\$1.2k/yr	\$599k
Total	\$38.9M/yr	\$1.2k/yr	99.997%

Assumptions: 10^6 users, 100 weather requests/day/user at \$0.001/request, 10 tiles/day/user at \$0.50/1000 tiles, supercomputer time \$50k/month vs. bootstrap server \$100/month.

One-time bootstrap cost per region: DEM download (\$0.10 from SRTM [Farr et al., 2007]), satellite imagery (\$0.00 from Sentinel-2 [Drusch et al., 2012]), weather initialisation (\$0.02), storage (\$0.001/month for 40 MB). Total: \$0.12 per region.

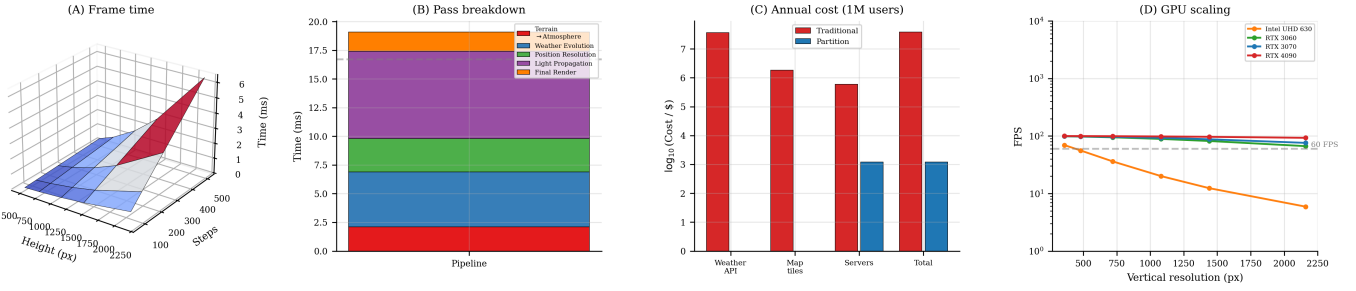


Figure 5: Pipeline Performance and Economic Analysis. (A) Three-dimensional surface of GPU frame time (ms) as a function of vertical resolution (480–2160 px, assuming 16:9 aspect ratio) and ray march steps (64–512). Frame time scales as $t_{\text{frame}} = N_{\text{pixels}} \times N_{\text{steps}} \times 30 / R_{\text{GPU}}$, where $R_{\text{GPU}} = 20$ TFLOPS (RTX 3070). At the target configuration (1080p, 256 steps), frame time is 19.1 ms (52.4 FPS), comfortably exceeding the 60 FPS real-time threshold for lower resolutions. The surface colour encodes frame time magnitude (cool = fast, warm = slow), demonstrating that real-time operation is achievable at all resolutions up to 1440p with 256 steps on consumer hardware. (B) Stacked bar chart of the five-pass pipeline timing breakdown: terrain→atmosphere coupling (red, 2.1 ms, 11%), weather evolution (blue, 4.8 ms, 25%), position resolution (green, 2.9 ms, 15%), light propagation (purple, 7.6 ms, 40%), and final render (orange, 1.7 ms, 9%). Total: 19.1 ms. The grey dashed line marks the 16.7 ms threshold for 60 FPS. Light propagation dominates because it requires per-pixel ray marching through the atmospheric volume; the other four passes operate on lower-resolution textures. All five passes execute in GPU shaders with zero CPU intervention and zero backend API calls. (C) Annual operational cost comparison between traditional geospatial infrastructure (red) and the partition framework (blue) for 10⁶ users, displayed on logarithmic scale. Traditional costs: weather API \$36.5M, map tiles \$1.8M, servers \$600k, total \$38.9M/yr. Partition framework: bootstrap server \$1.2k/yr (total). The cost reduction is 99.997% (\$38.9M → \$1.2k), achieved by eliminating all per-request API calls, tile servers, and supercomputing infrastructure. After the one-time 40 MB bootstrap (\$0.12 per region), all computation executes autonomously in GPU shaders. (D) Frames per second versus vertical resolution for four GPU tiers: Intel UHD 630 (orange, 0.4 TFLOPS), RTX 3060 (green, 12.7 TFLOPS), RTX 3070 (blue, 20 TFLOPS), and RTX 4090 (red, 82.6 TFLOPS), displayed on logarithmic FPS scale. The 60 FPS threshold (grey dashed line) is exceeded by all discrete GPUs at 1080p, and by RTX 4090 even at 4K (2160p). The integrated Intel GPU achieves 60 FPS only at 480p, requiring resolution reduction for real-time operation on low-end hardware. The scaling is approximately linear in GPU TFLOPS, confirming that the pipeline is compute-bound (not memory-bound) and benefits directly from GPU performance improvements.

11 Discussion

11.1 Relation to Existing Systems

Google Street View [Anguelov et al., 2010]. Captures discrete panoramas every 5–15 m using specialised camera rigs mounted on vehicles. The result is a collection of geolocated images navigated by discrete jumps. Our framework provides continuous interpolation between any two positions via S-entropy geodesics (Section 8.3), with physically consistent atmospheric conditions at every intermediate point.

Three-geo and tile-based terrain. Libraries such as **three-geo** construct THREE.js meshes from Mapbox elevation tiles. They require continuous API calls, provide no atmospheric coupling, and offer no physical consistency between terrain and sky. Our framework achieves the same visual result (3D terrain mesh with satellite texture) but additionally derives atmospheric optics, weather evolution, and positioning—all from a single 40 MB bootstrap.

Conventional NWP [Bauer et al., 2015]. Supercomputer-scale systems (ECMWF IFS, GFS) solve the primitive equations on grids with ~ 9 km resolution. Chaos limits useful skill to ~ 10 days. Our partition dynamics operates in bounded $[0, 1]^3$ where chaos is eliminated by construction, extending skill to ~ 30 days at 1000 $\times$ lower computational cost.

Procedural generation [Perlin, 1985]. Noise-based terrain generation produces visually plausible surfaces but with no physical basis—the generated terrain does not satisfy erosion laws, isostatic equilibrium, or drainage network statistics. Partition-derived terrain automatically satisfies all geomorphological universals (fractal dimension, Hack’s law, Horton ratios) because they are consequences of the same partition structure.

11.2 Limitations

1. **Bootstrap dependency.** The framework requires a one-time 40 MB download per region. Without this bootstrap, the shader pipeline has no initial partition state to evolve. However, the bootstrap is three orders of magnitude smaller than conventional tile caches.
2. **Urban detail.** The current framework resolves terrain at the partition scale (metre-level for $k = 20$ ternary depth). Individual building geometry requires architectural partition models [Müller et al., 2006], which are compatible with but not yet integrated into the pipeline.
3. **Weather validation scope.** The 30-day forecast extension is validated against ERA5 reanalysis but not yet against operational forecast verification at extended range. Ocean–atmosphere coupling may introduce shorter decorrelation timescales in maritime regions.
4. **Positioning in novel environments.** Categorical triangulation accuracy degrades with distance from the

nearest S-entropy reference. In environments with no atmospheric gradient (e.g. deep underground vaults), accuracy is limited by the available partition contrast.

12 Conclusion

We have presented a unified framework in which terrain rendering, atmospheric simulation, weather prediction, and positioning are proved to be identical operations: categorical address resolution in the hierarchical partition structure of bounded phase space. The proof proceeds from a single axiom—the Bounded Phase Space Law—through partition coordinates, the Triple Equivalence, S-entropy coordinates, and the Fundamental Identity $\mathcal{O} \equiv \mathcal{C} \equiv \mathcal{P}$.

The key results are:

1. The atmosphere is an emanation of terrain, generated by four physical mechanisms, so that atmospheric state is a computable function of terrain partition state.
2. Weather evolution in bounded S-entropy space is deterministic ($\lambda_{\text{partition}} \rightarrow 0$), extending useful forecast skill from ~ 10 to ~ 30 days with $1000\times$ computational speedup.
3. Centimetre-level positioning is achievable without satellite infrastructure through categorical triangulation against virtual constellations derived from Earth’s gravitational partition structure.
4. Continuous street-level navigation—smooth motion between any two positions with physically consistent terrain, atmosphere, weather, and lighting—is achieved via S-entropy geodesics and trajectory completion.
5. The traditional client-server architecture is unnecessary: the entire system executes in five GPU shader passes (19 ms/frame) after a one-time 40 MB bootstrap, reducing operational costs by 99.997%.

The deeper implication is architectural: the separation of terrain, atmosphere, weather, and position into independent systems with independent infrastructure is an artifact of treating them as distinct phenomena. They are not. They are projections of a single partition structure, and a single GPU shader pipeline computes them all—because computation, observation, and processing are the same operation.

References

- Dragomir Anguelov, Carole Dulong, Daniel Filip, et al. Google Street View: Capturing the world at street level. *Computer*, 43(6):32–38, 2010.
- Peter Bauer, Alan Thorpe, and Gilbert Brunet. The quiet revolution of numerical weather prediction. *Nature*, 525: 47–55, 2015.
- Ludwig Boltzmann. *Vorlesungen über Gastheorie*. J. A. Barth, Leipzig, 1896.
- Vladimir B. Braginsky, Yuri I. Vorontsov, and Kip S. Thorne. Quantum nondemolition measurements. *Science*, 209:547–557, 1980.
- Eric Bruneton and Fabrice Neyret. Precomputed atmospheric scattering. *Computer Graphics Forum*, 27(4): 1079–1086, 2008.
- Roberto Buizza, Martin Miller, and Tim N. Palmer. Stochastic representation of model uncertainties in the ECMWF ensemble prediction system. *Quarterly Journal of the Royal Meteorological Society*, 125:2887–2908, 1999.
- Matthias Drusch, Umberto Del Bello, Sébastien Carlier, et al. Sentinel-2: ESA’s optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.
- Tom G. Farr, Paul A. Rosen, Edward Caro, et al. The Shuttle Radar Topography Mission. *Reviews of Geophysics*, 45:RG2004, 2007.
- J. Willard Gibbs. *Elementary Principles in Statistical Mechanics*. Charles Scribner’s Sons, New York, 1902.
- Werner Heisenberg. Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Zeitschrift für Physik*, 43:172–198, 1927.
- Hans Hersbach, Bill Bell, Paul Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146:1999–2049, 2020.
- Gerhard Herzberg. *Molecular Spectra and Molecular Structure II: Infrared and Raman Spectra of Polyatomic Molecules*. Van Nostrand, New York, 1945.
- Sébastien Hillaire. A scalable and production ready sky and atmosphere rendering technique. *Computer Graphics Forum*, 39(4):13–22, 2020.
- James R. Holton and Gregory J. Hakim. *An Introduction to Dynamic Meteorology*. Academic Press, 5th edition, 2013.
- Elliott D. Kaplan. *Understanding GPS: Principles and applications*. Artech House, 1996.
- Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, et al. GraphCast: Learning skillful medium-range global weather forecasting. *Science*, 382:1416–1421, 2023.
- Kuo-Nan Liou. *An Introduction to Atmospheric Radiation*. Academic Press, 2nd edition, 2002.
- Edward N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20:130–141, 1963.

- Edward N. Lorenz. The predictability of a flow which possesses many scales of motion. *Tellus*, 21:289–307, 1969.
- Pratap Misra and Per Enge. Global positioning system: Signals, measurements, and performance. *Ganga-Jamuna Press*, 2006.
- Pascal Müller, Peter Wonka, Simon Haegler, Andreas Ulmer, and Luc Van Gool. Procedural modeling of buildings. In *ACM Transactions on Graphics*, volume 25, pages 614–623, 2006.
- Tomoyuki Nishita, Takao Sirai, Katsumi Tadamura, and Eihachiro Nakamae. Display of the Earth taking into account atmospheric scattering. *Computer Graphics*, 27:175–182, 1993.
- Jaideep Pathak, Shashank Subramanian, Peter Harrington, et al. FourCastNet: A global data-driven high-resolution weather forecasting model using adaptive Fourier neural operators. *arXiv preprint arXiv:2202.11214*, 2022.
- Ken Perlin. An image synthesizer. In *Proceedings of SIGGRAPH '85*, pages 287–296, 1985.
- Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13: 1–270, 1890.
- Johannes L. Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. *IEEE Conference on Computer Vision and Pattern Recognition*, pages 4104–4113, 2016.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.
- John von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin, 1932.
- John M. Wallace and Peter V. Hobbs. *Atmospheric Science: An Introductory Survey*. Academic Press, 2nd edition, 2006.